50 Most Important Deep Learning Viva Questions

Basic Definitions

1. **What is Artificial Intelligence (AI)?** --- AI is the field of building systems that perform tasks requiring human-like intelligence.
2. **What is Machine Learning (ML)?** --- ML enables computers to learn patterns from data.
3. **What is Deep Learning?** --- Deep learning uses neural networks with multiple layers to learn patterns.
4. **What is an Artificial Neural Network (ANN)?** --- An ANN is a network of connected units that learn by adjusting weights and biases.
5. **What is a neuron?** --- A neuron calculates a weighted sum of inputs and applies an activation function.
6. **What are weights and biases?** --- Weights control input influence, while biases provide trainable offsets.
7. **What is an epoch?** --- One epoch is one complete pass through the training dataset.
8. **What is batch size?** --- Batch size is the number of examples processed before one parameter update.
9. **What is a learning rate?** --- The learning rate controls the size of parameter updates.
10. **What is an optimizer?** --- An optimizer updates model parameters to reduce the loss.
11. **What is a loss function?** --- A loss function measures the difference between predictions and target values.
12. **What is backpropagation?** --- Backpropagation computes gradients used to update neural-network parameters.
13. **What is an activation function?** --- An activation function transforms a neuron's input and introduces nonlinearity.
14. **What is ReLU?** --- ReLU returns the input if positive and zero otherwise.
15. **What is softmax?** --- Softmax converts class scores into probabilities that sum to one.
16. **What is overfitting?** --- Overfitting occurs when a model fits training data well but performs poorly on unseen data.
17. **What is underfitting?** --- Underfitting occurs when a model fails to learn the patterns in the data.

18. **What is the difference between training and testing?** --- Training updates model parameters, while testing evaluates the trained model on unseen data.

Datasets and Preprocessing

19. **What is MNIST?** --- MNIST is a dataset of handwritten digits from 0 to 9.
20. **What is the size of an MNIST image?** --- Each MNIST image is 28 × 28 pixels and grayscale.
21. **How many images are in MNIST?** --- MNIST contains 60,000 training images and 10,000 test images.
22. **What is CIFAR-10?** --- CIFAR-10 is a dataset of small color images in ten object classes.
23. **What is the size of a CIFAR-10 image?** --- Each image is 32 × 32 pixels with three color channels.
24. **Why divide image pixels by 255?** --- It normalizes pixel values from 0--255 to approximately 0--1.
25. **What is one-hot encoding?** --- It represents a class as a vector with 1 at its class index and 0 elsewhere.
26. **Why reshape MNIST images to (28, 28, 1)?** --- CNNs expect image height, width, and channel dimensions.
27. **What does `mnist.load_data()` do?** --- It loads the MNIST training and test images and labels, downloading them if needed.

CNN and VGG19

28. **What is a CNN?** --- A convolutional neural network learns image features using convolutional filters.
29. **What is a convolutional layer?** --- It applies learnable filters to an input to produce feature maps.
30. **What is a filter or kernel?** --- A kernel is a small set of weights that detects patterns in an image.
31. **What is pooling?** --- Pooling reduces the spatial dimensions of feature maps.
32. **What is max pooling?** --- Max pooling selects the maximum value from each local window.
33. **What does a Flatten layer do?** --- It converts multidimensional feature maps into a one-dimensional vector.

34. **What is VGG19?** --- VGG19 is a convolutional neural network architecture with 19 weight layers.
35. **What is transfer learning?** --- Transfer learning reuses a pretrained model for a new task.
36. **What does `weights='imagenet'` mean?** --- It loads weights pretrained on the ImageNet dataset.
37. **What does `include_top=False` mean?** --- It removes VGG19's original final classification layers.
38. **Why freeze VGG19 layers?** --- Freezing preserves pretrained weights while training the new classification head.
39. **What is dropout?** --- Dropout randomly disables units during training to help reduce overfitting.

Gradient Descent and Optimizers

40. **What is gradient descent?** --- Gradient descent updates parameters in the direction that reduces loss.
41. **What is SGD?** --- Stochastic gradient descent updates parameters using gradients estimated from a batch of examples.
42. **What is momentum?** --- Momentum uses past gradients to smooth and accelerate parameter updates.
43. **What is Nesterov momentum?** --- Nesterov momentum uses a look-ahead gradient to adjust the update direction.
44. **What is Adam?** --- Adam adapts parameter updates using moving averages of gradients and squared gradients.
45. **What is RMSprop?** --- RMSprop scales updates using a moving average of squared gradients.
46. **What is Adagrad?** --- Adagrad adapts learning rates using accumulated squared gradients.
47. **Which optimizer is always best?** --- No optimizer is always best; performance depends on the model, data, and hyperparameters.

TensorFlow and Evaluation

48. **What is TensorFlow?** --- TensorFlow is a framework for numerical computation and machine learning.
49. **What does `model.fit()` do?** --- It trains the model on the supplied data.

50. **What is accuracy?** --- Accuracy is the number of correct predictions divided by the total number of predictions.